Compiler Design

Lecture-06

LL(1) Parser

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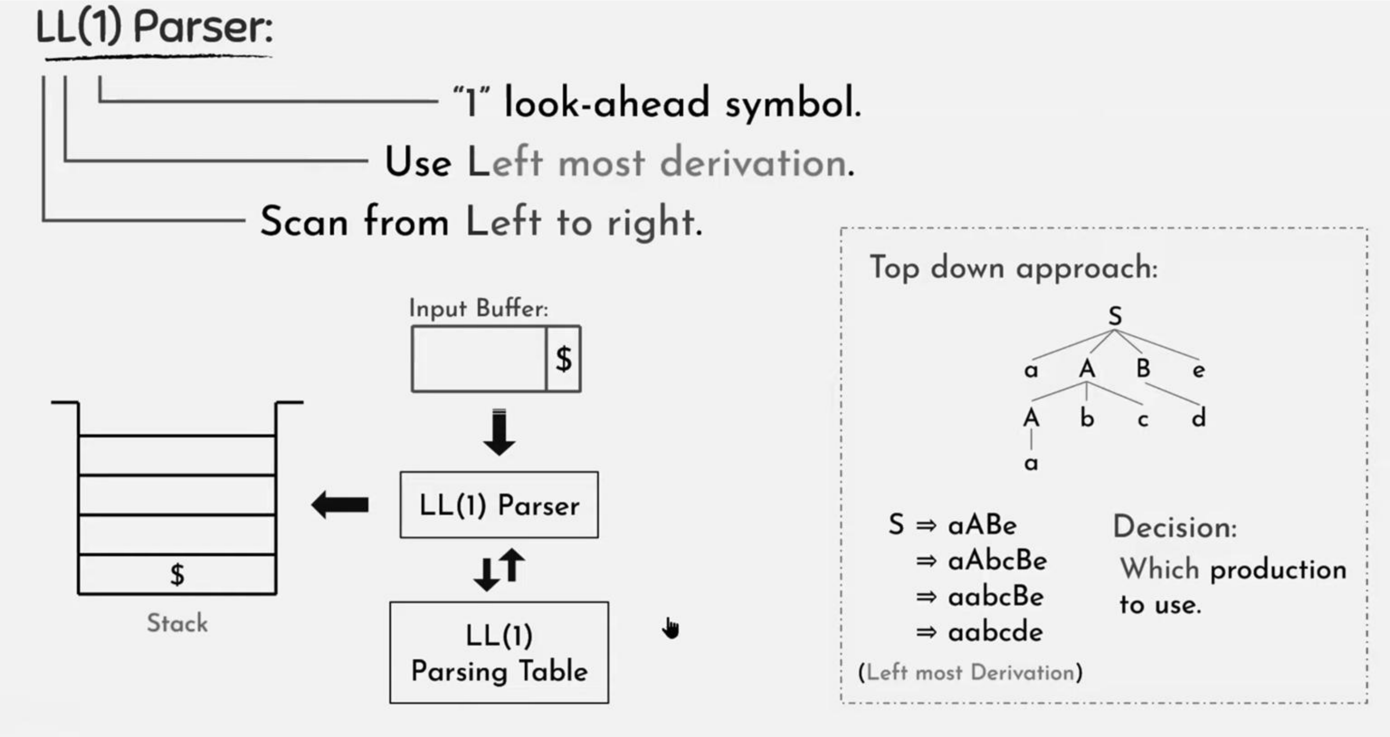
Course: CSE 4111

LL(1) Parser


LL(1) Parser:
"1" look-ahead symbol.
Use Left most derivation.
Scan from Left to right.
Input Buffer:
$
LL(1) Parser
LL(1) Parsing Table
$
Stack
Top down approach:
S
a A B e
A b c d
a
S ⇒ aABe
⇒ aAbcBe
⇒ aabcBe
⇒ aabcde
(Left most Derivation)
Decision:
Which production to use.

First()

LL(1) Parser:

1. FIRST():

   Given any non-terminal of a CFG, if we derive all the possible strings from it, the first terminal(s) is the FIRST() of the non-terminal.

e.g.(1):
$S \rightarrow aABC$
$A \rightarrow b$
$B \rightarrow c$
$C \rightarrow d$

FIRST(S): S → a A B C

FIRST(A): A → b

FIRST(B): B → c

FIRST(C): C → d

First() cont'd

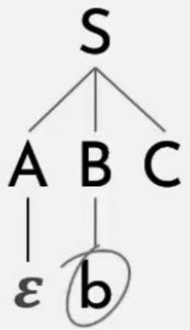
e.g.(2): $S \rightarrow ABC$

$A \rightarrow a \mid \varepsilon$

$B \rightarrow b$

$C \rightarrow c$

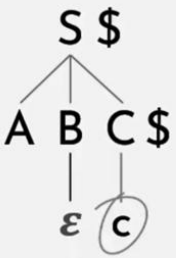
FIRST(S): { a, b }



Follow()

During the process of derivation, the terminal(s) that could follow the non-terminal are to be considered as FOLLOW() of the non-terminal.

e.g.: $S \rightarrow ABC$
$A \rightarrow a$
$B \rightarrow b \mid \varepsilon$
$C \rightarrow c$



FOLLOW(S): { $ }
FOLLOW(A): { b, c }
FOLLOW(B): { c }
FOLLOW(C): { $ }

- **In Follow() epsilon isn't included.**

Rules for Follow()

1) FOLLOW(S) = { $ } // where S is the starting Non-Terminal

2) If A -> pBq is a production, where p, B and q are any grammar symbols, then everything in FIRST(q) except Є is in FOLLOW(B).

3) If A->pB is a production, then everything in FOLLOW(A) is in FOLLOW(B).

4) If A->pBq is a production and FIRST(q) contains Є, then FOLLOW(B) contains { FIRST(q) – Є } U FOLLOW(A)

Derivation of FIRST:

$E \rightarrow TE'$
$E' \rightarrow +TE' \mid \varepsilon$
$T \rightarrow FT'$
$T' \rightarrow *FT' \mid \varepsilon$
$F \rightarrow id \mid (E)$

	FIRST
$E \rightarrow TE'$	{id,(}
$E' \rightarrow +TE' \mid \varepsilon$	{+,ε}
$T \rightarrow FT'$	{id,(}
$T' \rightarrow *FT' \mid \varepsilon$	{*,ε}
$F \rightarrow id \mid (E)$	{id,(}

FIRST(F) = {id,(}
FIRST(T') = {*,ε}
FIRST(T) = {id,(}
FIRST(E') = {+,ε}
FIRST(E) = {id,(}

Derivation of FOLLOW:

	FIRST	FOLLOW
$E \rightarrow TE'$	{id,(}	{\$,)}
$E' \rightarrow +TE' \mid \varepsilon$	$\{+,\varepsilon\}$	{\$,)}
$T \rightarrow FT'$	{id,(}	{+
$T' \rightarrow *FT' \mid \varepsilon$	$\{*,\varepsilon\}$	
$F \rightarrow id \mid (E)$	{id,(}	

$$FOLLOW(T) = FIRST(E') = \{ +, \varepsilon \}$$

1. The following terminal symbol will be selected as FOLLOW.
2. The FIRST of the following non-terminal will be selected as FOLLOW.
3. If it is the right most in the RHS, the FOLLOW of the LHS will be selected.

	FIRST	FOLLOW
$E \rightarrow TE'$	{id,(}	{$,)}
$E' \rightarrow +TE' \mid \varepsilon$	{+,ε}	{$,)}
$T \rightarrow FT'$	{id,(}	{+,$,)}
$T' \rightarrow *FT' \mid \varepsilon$	{*,ε}	{+,$,)}
$F \rightarrow id \mid (E)$	{id,(}	{*,+,$,)}

Assignment

Q1.

Find the FIRST and FOLLOW of all the non-terminals:

$$S \rightarrow ABCDE$$
$$A \rightarrow a \mid \varepsilon$$
$$B \rightarrow b \mid \varepsilon$$
$$C \rightarrow c$$
$$D \rightarrow d \mid \varepsilon$$
$$E \rightarrow e \mid \varepsilon$$

Q2. Find First() and Follow()

S-> Bb|Cd

B-> aB| ε

C-> cC| ε

Construction of LL(1) Parsing Table

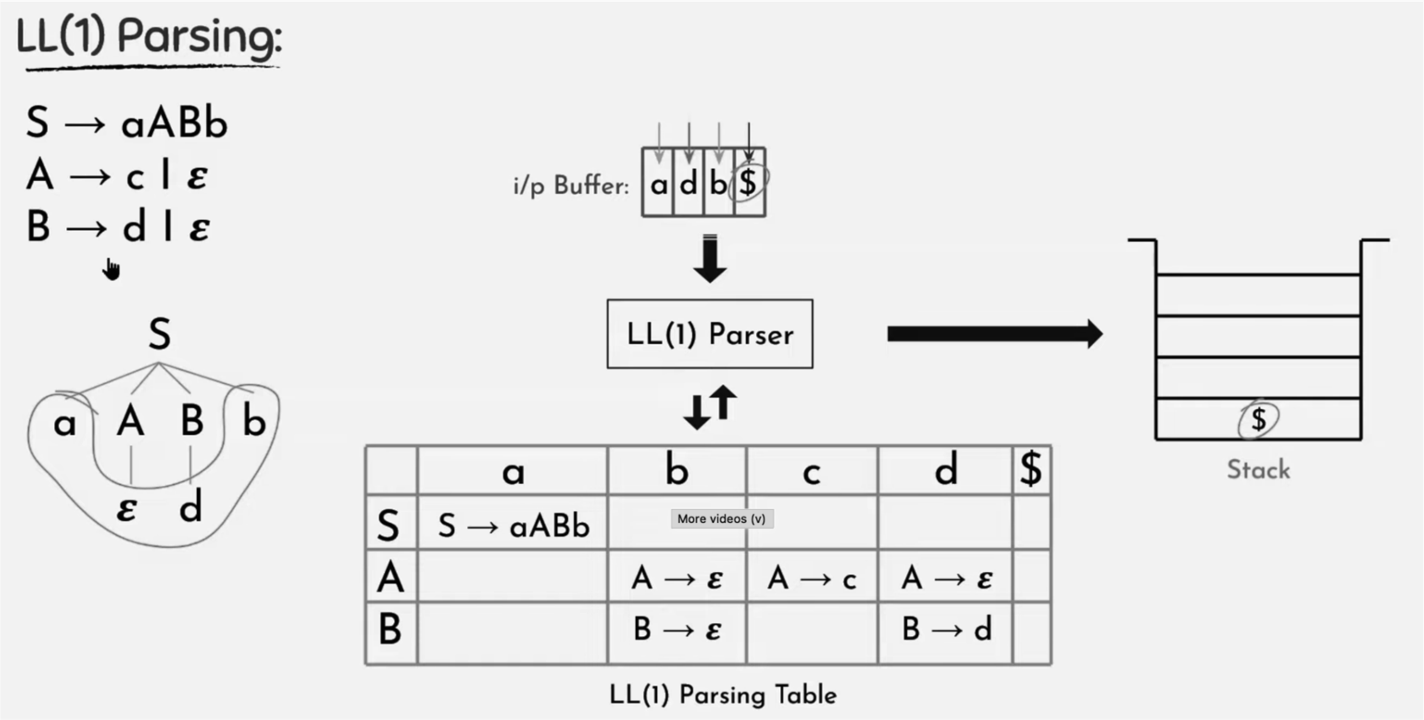
	id	(	)	*	+	$
E	$E \to TE'$	$E \to TE'$				
E'			$E' \to \varepsilon$		$E' \to +TE'$	$E' \to \varepsilon$
T	$T \to FT'$	$T \to FT'$				
T'			$T' \to \varepsilon$	$T' \to *FT'$	$T' \to \varepsilon$	$T' \to \varepsilon$
F	$F \to id$	$F \to (E)$				

Rules:

1. All the ε-productions are placed under FOLLOW sets.
2. Remaining productions are placed under the FIRSTs.

	FIRST	FOLLOW
$E \to TE'$	{id,(}	{$,)}
$E' \to +TE' \mid \varepsilon$	{+,ε}	{$,)}
$T \to FT'$	{id,(}	{+,$,)}
$T' \to *FT' \mid \varepsilon$	{*,ε}	{+,$,)}
$F \to id \mid (E)$	{id,(}	{*,+,$,)}

LL(1) Parsing



	a	b	c	d	$
S	$S \to aABb$				
A		$A \to \varepsilon$	$A \to c$	$A \to \varepsilon$	
B		$B \to \varepsilon$		$B \to d$	

LL(1) Parsing Table

Find out whether the following grammar is LL(1):

$$S \rightarrow aSA \mid \varepsilon$$
$$A \rightarrow c \mid \varepsilon$$

	FIRST	FOLLOW
$S \rightarrow aSA \mid \varepsilon$	$\{a, \varepsilon\}$	$\{\$, c\}$
$A \rightarrow c \mid \varepsilon$	$\{c, \varepsilon\}$	$\{\$, c\}$

	a	c	\$
S	$S \rightarrow aSA$	$S \rightarrow \varepsilon$	$S \rightarrow \varepsilon$
A		$A \rightarrow c$ $A \rightarrow \varepsilon$	$A \rightarrow \varepsilon$

Find out whether the following grammar is LL(1):

$$S \rightarrow A$$
$$A \rightarrow Bb \mid Cd$$
$$B \rightarrow aB \mid \varepsilon$$
$$C \rightarrow cC \mid \varepsilon$$

	FIRST	FOLLOW
$S \to A$	{a, b, c, d}	{\$}
$A \to Bb \mid Cd$	{a, b, c, d}	{\$}
$B \to aB \mid \varepsilon$	{a, ε}	{b}
$C \to cC \mid \varepsilon$	{c, ε}	{d}

	a	b	c	d	\$
S	$S \to A$	$S \to A$	$S \to A$	$S \to A$	
A	$A \to Bb$	$A \to Bb$	$A \to Cd$	$A \to Cd$	
B	$B \to aB$	$B \to \varepsilon$			
C			$C \to cC$	$C \to \varepsilon$	

- **It's a LL(1) Grammar.**